Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/11/2013
PI Name:	Eisch, Amelia Ph.D.		
Project Title:	Molecular and Cellular Mechanisms of Space Radiation-Induced Changes in Hippocampal-Dependent Behavior		
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PI Email:	eischa@email.chop.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	(215) 590-1931
Organization Name:	Children's Hospital of Philadelphia/Univ Pennsylvania Perelman School of Medicine		
PI Address 1:	3401 Civic Center Blvd.		
PI Address 2:			
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104	**Congressional District:**	3
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Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
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	An unavoidable consequence of many space missions is exposure to galactic cosmic radiation (GCR), which includes HZE particles like Fe, Si, and O. Estimating radiation risks to the central nervous system (CNS) is a high research priority. Here we propose extensive analysis of how a specific region of the CNS, the hippocampus, is influenced by radiation. The hippocampus is important because it contributes to spatial learning and mood regulation, which are critical for mission success. Recently the hippocampal dentate gyrus (DG) was shown to be involved in learning challenging tasks requiring fine spatial distinction ("pattern separation"). The hippocampus is also important because it is exceptionally "neuroplastic". For example, in terms of gene transcription changes, the DG is among the most responsive CNS regions to physiological and environmental stimuli. The DG is also a site of neurogenesis throughout life, and adult neurogenesis is proposed to contribute to hippocampal function in general and pattern separation in particular.		

Task Description:	Understanding how space radiation influences DG structure and function will allow far better estimation of CNS radiation risks. Research already suggests that space radiation has a negative effect on hippocampal structure and function. However, it is unknown whether space radiation specifically influences DG functions like pattern separation. It is also unknown whether the diverse neural cells types within the DG are similarly influenced by space radiation or whether, for example, there is a cell-type specific change in translation or morphology. We will use recent advances in knowledge of DG function and structure and cutting-edge but proven technical approaches to answer key remaining questions about how space radiation influences the cellular, molecular, and functional integrity of the hippocampus. For example in Aim 1, using behavioral tests with which we have recently published (delayed non-match to pattern radial arm maze for fine spatial distribution; Guo et al., Nat Med 2011), we will explore our hypothesis that space radiation decreases pattern separation in a dose- and LET-dependent manner. In Aim 2, using transgenic mice we have developed that allow inducible YFP labeling of neural stem cells and their progeny (nestin-CreERT2/R26R mice; Lagace et al., J Neurosci 2007), we will explore our hypothesis that space radiation negatively impacts early postnatally-generated DG neurons. In Aim 3, using transgenic mice developed by our colleagues that allow profiling of translating RNA in genetically-defined DG neural cells, we will explore our hypothesis that space radiation causes discrete changes in DG neural cell types in a dose- and LET-dependent manner. Unlike currently used approaches to assess total RNA profiles from cellularly heterogeneous samples (e.g. laser capture dissection and subsequent microarray analysis), this approach allows purification of specific cell types via ribosome tag and a quantitative comparison of translating RNA profiles in genetically-distinct DG cells. Importantly, all mice will be 6-7 month old at time of radiation exposure in order to be approximately the age equivalent to a 30 year old human, and thus to better assess the impact of ground-based space radiation on the fully mature brain. Tissue from Aims 1-3 will be examined using immunohistochemistry, epifluorescent and confocal microscopy, stereologic cell counting, and morphological analysis, and tissue from Aim 3 will be explored using microarrays and RNAseq on the translating RNA pools and qRT-PCR. Taken together, these aims will allow us to test our overall hypothesis that radiation will have distinct, long-lasting effects on the behavioral, cellular, and molecular integrity of the hippocampus. Thus, these results will expedite the quest to estimate CNS radiation risks due to GCR exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is critical to identify factors that regulate normal brain function and structure. Our proposed research will elucidate how ground-based space radiation influences hippocampal structure and function. The knowledge derived from the proposed experiments will provide insight into the mechanisms contributing to brain and hippocampal function and their disruption by space radiation, and thus has the potential to shed light on how we can estimate CNS risk from space radiation. This work is important given the long-duration space missions on the former Mir Space Station, the present International Space Station, and the potential exploratory missions to the Moon and Mars. Mice will be used as the primary source of nervous tissue for experimentation since most of the relevant literature concerns studies of tissues of rodent origin, and since the mouse is an excellent model for human exposure to space radiation given their remarkably similar brain structure and function and genomic similarity. At this time, using computer modeling to understand the effects of space radiation on behavioral, cellular, and molecular indices of the hippocampus is just not possible given the complexity of the CNS. Cell culture systems have proven extremely useful to ask mechanistic questions about the effects of space radiation on the hippocampus, but they are unable to address the effects of space radiation on the whole organism, particularly behavior. The combination of approaches used here - analysis of behavior, cellular morphology, and translational profile - will have significant benefits for the fields of neuroscience in general and the fields of learning and memory, stem cell biology, and even neural development in particular, but are all inspired by the need to gain greater insight into whether the CNS alterations experienced by astronauts may compromise mission success. The particular focus of this application - hippocampal dysfunction - is evident in a range of psychiatric and neurologic disorders, including Alzheimer’s Disease, depression, and addiction. Astronauts may be particularly at risk for delayed hippocampal pathological signs, as it is estimated that even just 1 year in a low-Earth orbit mission may result in 91% of hippocampal cells and 25% of hippocampal cell-nuclei hit by high LET particles. These studies will help us predict neurodegenerative or cognitive changes in astronauts during the mission (which would compromise mission success) or even after mission completion. We also focus on the hippocampus since it is exceptionally "neuroplastic" – being among the most responsive brain regions in terms of gene transcription after physiological manipulation and stimuli and being a site of the birth of new neurons throughout life. This exceptional neuroplasticity suggests that the hippocampus may be among the first brain regions to reveal a response to radiation. To best study the influence of ground-based space radiation on hippocampal structure and function in a way that is most translational from rodents to humans, we will be specifically using older mice (e.g. ~6-7months old at time of irradiation, which is equivalent to 30 human years). Aside from the general benefits of this broader work for neuroscience and humans, there are several innovative aspects to this work. For example, using mice that we have generated that allow inducible labeling and tracking of embryonic vs. adult-generated neurons, we will be able to for the first time tease apart effects on these two populations. In addition, using novel mice that allow profiling of translating RNA in genetically-defined DG neural cells (e.g. adult-generated vs. embryonic-generated DG neurons), we will make a quantitative comparison of translating RNA profiles in genetically-distinct DG neural cells in a time-, dose-, and LET-dependent manner. Utilization of these novel technologies have the implication to expand our broader understanding of mechanisms underlying neuroplasticity and even neurodegenerative processes.

Task Progress:

Summary of progress through Year 2 (11/2012 – 11/2013) of 3 years of funding.
As highlighted in the numbered findings below each of our 3 specific aims, we have met or exceeded all goals for this 2nd year.

Aim 1: Determine the impact of low- and high-LET radiation on hippocampal function.

1. Building off of the first year's work (NSRL 11C and NSRL 12B), we developed a robust hippocampal-dependent task, called context discrimination fear conditioning (CDFC), to assess a subtle but important hippocampal ability called "pattern separation" in mice. Critically, we are consistently able to assess pattern separation in both young and aged mice, positioning us to assess whether HZE particles influence hippocampal function in aged mice that are the rough equivalent age of human astronauts.

2. We have used CDFC to test the effects of 28Si and 56Fe irradiation on hippocampal function in aged mice (NSRL 12C & 13A). In contrast to our original hypothesis, we found that aged irradiated mice have improved pattern separation (a subtle but critical hippocampal function) relative to controls.

a. For example, in NSRL 12C (28Si, 275 MeV/n) ~7 month-old mice irradiated with 100 cGy and tested at ~11 months of age show earlier acquisition in discriminating between two contexts that look different but share similar features. Furthermore, mice that received the higher irradiated dose (100 cGy) showed more consistent discrimination than the lower dose (20cGy) or control mice over the 16-day CDFC paradigm.

b. Similarly, in NSRL 13A (fractionated and non-fractionated 56Fe irradiation, 600 MeV/n), aged mice showed enhanced pattern separation compared to non-irradiated controls. Both fractionated and non-fractionated irradiated mice showed earlier and more consistent context discrimination learning (pattern separation) than non-irradiated controls, with fractionated showing earlier discrimination learning than non-fractionated.

c. While both 28Si and 56Fe irradiation improve pattern separation in aged mice, neither influence other behaviors (e.g. locomotion, exploratory of a novel environment, basal levels of anxiety-like behavior, pain threshold).

d. We are currently testing the effects of 28Si in young mice (NSRL 13B, ~9 weeks of age at irradiation) to see if this is an age-specific effect (compare to NSRL 12C). Mice are being analyzed (Sept-Dec 2013) for behavioral parameters 3-6 months post-irradiation. We hypothesize that young mice exposed to 28Si irradiation will show deficits in pattern separation.

3. We have also identified an age-dependent influence of HZE particle irradiation on hippocampal-dependent function. Specifically, we used context dependent fear conditioning (CFC) to test the effects of 28Si irradiation on hippocampal function in young and aged mice (NSRL 13B and 12C). Young mice show a dose-dependent influence of HZE particle radiation on CFC, with 20 cGy causing hippocampal-dependent deficits and 100 cGy causing general anxiety. Similar results have been seen with X-ray radiation, but this has not been observed yet in mice after HZE particle irradiation. Even more interestingly, older mice show no effect of HZE particle radiation on CFC performance. We are eager to explore whether hippocampal neurogenesis is differentially influenced by radiation in young vs. older mice. This will be a focus of Year 3.

4. Behavior testing has also revealed a potential interaction of radiation and behavioral stress. For example, in NSRL 13A (56Fe), non-fractionated irradiated mice have greater anxiety (exploration of a novel environment) at both a short and long delay between anxiety testing and a shock stress relative to non-fractionated irradiated mice that did not experience the shock stress. This effect was not seen in fractionated mice, and was only seen in control mice when the delay between anxiety testing and the shock was short. This suggests that stressful experience may alter the long-term influence of radiation on behavior. In addition, how the radiation was delivered (fractionated vs. non-fractionated) also may play a role in the influence of radiation on behavior. This will be a focus of Year 3 studies.

5. In response to previous aggression issues in our mice (see last progress report), we have made modifications to our transport and housing procedures that have significantly diminished aggression in our aged mice. Specifically, we have utilized housing enrichment and lower packing density. Notably, other NASA-funded investigators have mentioned they have similar problems with aggression in their aged animals (Richard Britten, Bernie Rabin, personal communication).

Aim 2: Determine how radiation influences the dendritic morphology of embryonic- vs. adult- generated granule cell neurons.

1. Using tissue from a pilot study (NSRL 13A), we made two important advances.

a. First, we confirmed our technical approach of labeling early postnatally-generated hippocampal neurons (via Nestin-CreERT2/R26R-YFP mice receiving Tamoxifen at P3) and visualizing them in adulthood (P90). In fact, we published a paper on a related approach in PLoS One this year (Petrik et al., 2013).

b. Second, we confirmed our hypothesis that there would be no gross alteration in number of YFP+ neurons in the dentate gyrus after 56Fe irradiation at 60 days of age (600 MeV/n; 20 cGy). We are currently assessing a key aspect of our initial hypothesis: that embryonic-generated hippocampal neurons in irradiated mice will display some dendritic disruption relative to controls. These data will be collected before NSRL 14A and will guide us in whether we need additional animals at 30d post-irradiation or whether we can use a new timepoint (likely 7 days post-irradiation).

2. In addition to exploring the influence of 56Fe irradiation on dendritic morphology of embryonic-generated dentate gyrus neurons, we have also established a cohort of mice to assess the impact of 28Si irradiation (275 MeV/n) on this as well (NSRL 13C). We are currently breeding and administering Tamoxifen to our Nestin-CreERT2/R26R-YFP mice at P3 of age. These mice are scheduled for 28Si irradiation during our upcoming fall run (October 29, 2013). Following irradiation, brain tissue will be harvested for histological analysis (likely 30d post-irradiation) to identify differences in dendritic morphology in embryonic-generated granule cell neurons among control and irradiated (20 cGy or 100 cGy) mice.

3. Finally we are breeding for two complementary studies to be assessed in NSRL 14A:

a. First, we will examine the impact of irradiation on dendritic morphology in embryonic-generated granule cell neurons in aged mice.

b. Second (should we receive the 4th year of funding), we will examine the impact of irradiation on dendritic morphology in adult-generated granule cell neurons in aged mice.

Aim 3: Define the impact of radiation on the translational profile of genetically defined DG neural cells.

1. We have expanded the transgenic mouse line needed for this aim to mouse numbers suitable for experimentation. Specifically, we are currently breeding our Cx3CL1-BACTRAP transgenic mice for NSRL 13C. We will keep this initial run smaller than usual (~n=20) in order to confirm that this mouse line is equally responsive to our efforts in housing and shipping that keep aggression at a low level in other lines. We will send a larger cohort in 2014 (NSRL 14A, 14B) for radiation exposure to build on these initial studies in NSRL 13C.

2. We have validated our immunoprecipation assays for harvesting and quantitating mRNA expression using non-irradiated Cx3CL1-BACTRAP mice. We have interviewed several of the core facilities on campus, and have decided which one to work with for our larger number of experimental samples. We will utilize this technique following HZE radiation exposure. Notably, we have opted to try both microarray and RNAseq in order to have an unbiased screen for which proteins are being differentially translated in the dentate gyrus granule cells of irradiated vs. control mice.

Additional progress:

1. As our behavioral progress advanced, we became more efficient in animal use, allowing us to assess other aspects of behavior with animals that would have gone to CFC or CDFC testing. One other aspect of behavior we assessed in NSRL 13A (56Fe irradiation) was circadian rhythm, as HZE particle irradiation may cause alterations in brain regions responsible for this rhythm. While non-fractioned irradiated mice had the same length of circadian cycle as control mice, a “jet lag” experiment where we shifted the timing of the light by 6h revealed the unexpected result that HZE particle irradiation leads to a trend for improved entrainment to a light shift (“jet lag paradigm”) in aged mice. While we are uncertain if we have the funds, we would ideally like to assess this in fractionated mice as well, or to perform other circadian analyses. We will discuss this with the NASA Project Scientist prior to any more tests so we can receive feedback on this unexpected research direction.

2. We were also fortunate to interact with another NASA-funded investigator, Greg Nelson, and receive tissues from mice that were tested as part of the last space shuttle mission. We examined the olfactory bulb neurogenesis in these mice, and found a very interesting influence of STS-135 space flight and environmental enrichment on this measure. This work is in the final stages of preparation for submission.

Publications

We are also making excellent progress on publications, as we already have three articles published, one under revision, several more in preparation. This progress was due in part to the prior grant shared between Dr. Eisch and her co-I on this grant, Dr. Benjamin Chen, who was PI on NNA07AP84G, which has given the research team significant experience with these experiments. Articles published this year:

a. Decarolis NA, Mechanic M, Petrik D, Carlton AF, Malhotra S, Ables JL, Bachoo R, Lagace DC, Eisch AJ. In vivo contribution of nestin- and GLAST-lineage cells to adult hippocampal neurogenesis. Hippocampus. Apr 4. doi: 10.1002/hipo.22130. [Epub ahead of print]. PMID 23554226, PMC 3718551

b. Petrik D, Yun S*, Latchney SE*, Kamrudin S, LeBlanc JA, Bibb JA, Eisch AJ. Early postnatal in vivo gliogenesis from nestin-lineage progenitors requires Cdk5. PLoS One, Aug 26;8(8):e72819. doi: 10.1371/journal.pone.0072819. PMID 23991155, PMC in progress. *authors contributed equally to this work.

c. Rivera PD*, Shih H-Y*, LeBlanc JA, Cole MG, Amaral WZ, Mukherjee S, Zhang S, Lucero MJ, DeCarolis NA, Chen BPC, Eisch AJ. Acute and Fractionated Exposure to High-LET 56Fe HZE Particle Radiation Both Result in Similar Long-Term Deficits in Adult Hippocampal Neurogenesis. Radiation Research. In press. *authors contributed equally to this work. Articles in revision or preparation:

d. DeCarolis NA, Ahn HF, Lagace DC, Ables JL, Redfield RL, Cole MG, Chen BPC, Eisch AJ. Running after irradiation ameliorates the proliferation deficit but not the nestin-expressing Type-1 cell deficit in the adult hippocampal subgranular zone. Under revision, Hippocampus.

e. DeCarolis NA, Ahn HF, Amaral WZ, Rivera PD, LeBlanc JA, Malhotra S, Mukherjee S, Chen BPC, Eisch AJ. Impact of high-LET 56Fe particle irradiation on adult hippocampal neural stem cells and neurogenesis in vivo. In preparation for submission to Radiation Research.

f. Latchney SE, Rivera PD, Nelson G, Eisch AJ. Counteracting consequences of low earth orbit flight experience on olfactory bulb neurogenesis. In preparation for submission to Journal of Applied Physiology.

3. We have also presented oral/poster presentations at four conferences this past year, including:

a. Annual Meeting of the Society for Neuroscience (October, 2012)

b. NASA 2013 Human Research Program Investigators’ Workshop (February, 2013)

c. The Heavy Ion in Therapy and Space Radiation Symposium (May, 2013)

d. International Symposium on Radiation Science 2013 (May, 2013)

e. The 59th Annual Radiation Research Society Meeting (September, 2013)*

*Dr. Eisch is giving an invited “topical review” presentation, as well as her lab member is presenting a poster.

Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Decarolis NA, Mechanic M, Petrik D, Carlton A, Ables JL, Malhotra S, Bachoo R, Götz M, Lagace DC, Eisch AJ. "In vivo contribution of nestin- and GLAST-lineage cells to adult hippocampal neurogenesis." Hippocampus. 2013 Aug;23(8):708-19. Epub 2013 May 17. http://dx.doi.org/10.1002/hipo.22130 ; PubMed PMID: 23554226 ; PMCID: PMC3732558 , Aug-2013

Articles in Peer-reviewed Journals	Petrik D, Yun S*, Latchney SE*, Kamrudin S, LeBlanc JA, Bibb JA, Eisch AJ. (*authors contributed equally to this work) "Early postnatal in vivo gliogenesis from nestin-lineage progenitors requires Cdk5." PLoS One. 2013 Aug 26;8(8):e72819. http://dx.doi.org/10.1371/journal.pone.0072819 ; PMID: 23991155 , Aug-2013
Articles in Peer-reviewed Journals	Eisch AJ, Petrik D. "Depression and hippocampal neurogenesis: a road to remission?" Science. 2012 Oct 5;338(6103):72-5. Review. http://dx.doi.org/10.1126/science.1222941 ; PubMed PMID: 23042885 , Oct-2012
Articles in Peer-reviewed Journals	Rivera PD*, Shih H-Y*, LeBlanc JA, Cole MG, Amaral WZ, Mukherjee S, Zhang S, Lucero MJ, DeCarolis NA, Chen BPC, Eisch AJ. (*authors contributed equally to this work.) "Acute and Fractionated Exposure to High-LET 56Fe HZE Particle Radiation Both Result in Similar Long-Term Deficits in Adult Hippocampal Neurogenesis." Radiation Research. In press. September 2013. , Sep-2013